

WASP-36b

R-1858

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_161120 · created 2026-08-03T09:04:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457712.9293 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.049
Transit depth (Rp/Rs)^2	2.47 +/- 1.53 [%]
Orbital Inclination (inc)	81.63 +/- 2.89
Airmass coefficient 1 (a1)	0.74 +/- 0.45
Airmass coefficient 2 (a2)	0.14 +/- 0.12
Scatter in the residuals of the lightcurve fit is	64.64 %
Best Comparison Star	#1 - [617, 63]
Optimal Aperture	2.88
Optimal Annulus	15.91
Transit Duration (day)	0.057 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.049 vs 0.1368 (deviation 0.41σ)
Duration fit vs published	0.057 d vs 0.0758 d (deviation 0.61σ)
Mid-time O–C	5.4 min
Depth SNR	3.2
Residual scatter	64.64 %

Light curve

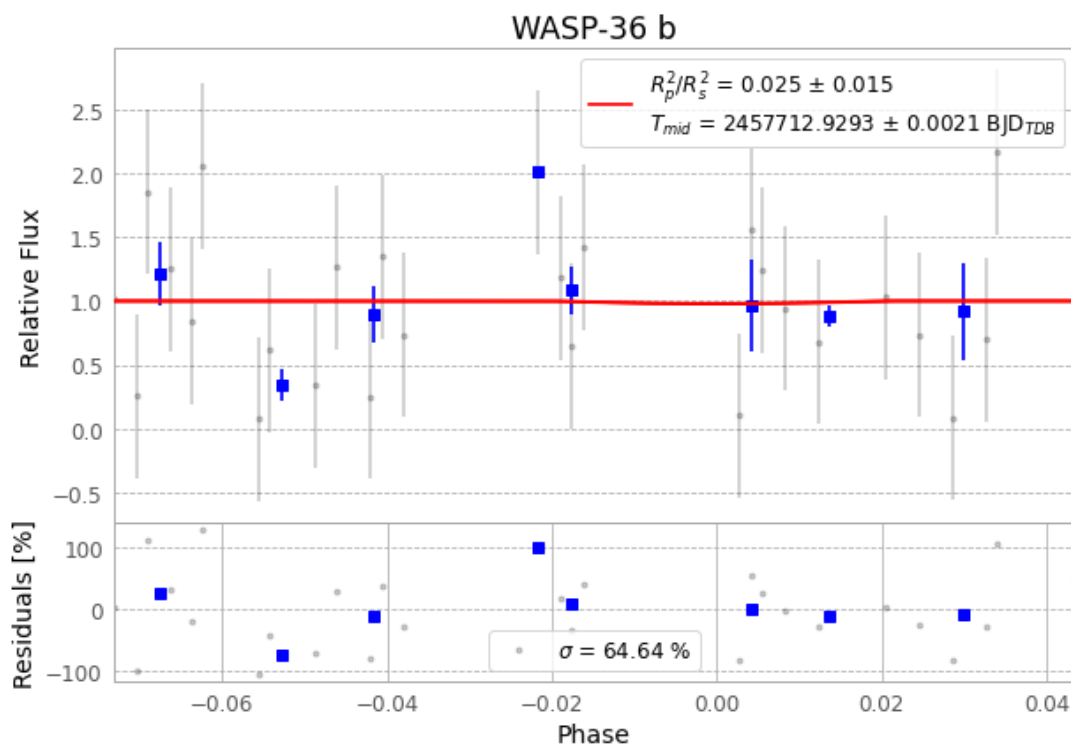


Figure 1 — FinalLightCurve_WASP-36 b_2016-11-20.png (SHA-256 8dc3b911305d4fdb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [618, 46], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6b53a215731c73e6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-36161120073012.FITS → WASP-36161120115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-161120.log (59633006306a...), FinalLightCurve_WASP-36 b_2016-11-20.png (8dc3b911305d...), FinalLightCurve_WASP-36 b_2016-11-20.csv (bb6ca79ca6b9...)

Compute resources

Wall time 285.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 09:04Z.